



Monkey see, monkey tap: mimicry of movement dynamics during coordinated tapping

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Abstract

Common social behaviors, such as having a conversation, dancing, or playing a team sport, require precise interpersonal coordination of action. One question that emerges in research on interpersonal coordination is to what extent individuals implicitly mimic the spatial characteristics of movements for tasks that emphasize movement timing. To investigate this question, we conducted two experiments using an interpersonal synchronization-continuation tapping paradigm in which pairs of individuals tapped with their index finger on a table in synchrony with an auditory metronome and then continued tapping at the same tempo when the metronome stopped. Pairs of individuals tapped either together with the instruction to maintain synchrony with each other (interpersonal tapping) or tapped alone (solo tapping). Solo tapping conditions either occurred with their tapping partner present in the testing room (Experiment 1) or absent (Experiment 2). We used motion capture to examine both the spatial and temporal aspects of movement dynamics during task performance. In both experiments, participants implicitly mimicked subtle aspects of spatial elements of their partner's movements. The extent of finger extension (tap amplitude) and, in Experiment 1, duration of finger contact with the surface (dwell time) were correlated between tapping partners when they tapped together. In some cases, this spatial mimicry extended to solo tapping conditions, but only during solo tapping conditions that followed the interpersonal tapping task, and, to a lesser degree, when solo tapping after having observed the other participant solo tapping.

Keywords Interpersonal synchronization · Mimicry · Finger tapping · Movement timing · Motion tracking

Introduction

Coordination of actions between individuals is a fundamental part of daily life that is central to most types of social interactions, such as carrying on a conversation, whether in person or on a video conference, dancing, playing a team sport, or more broadly, accomplishing almost any shared task or activity. When people interact in social contexts, there is accumulating evidence that individuals demonstrate

an implicit tendency to imitate each other's posture, gestures, or even characteristics of their speech (Bernieri and Rosenthal 1991; Chartrand and Bargh 1999; Schefflen 1964). Chartrand and Bargh (1999) found that when taking turns describing pictures with an unidentified confederate, naïve participants unconsciously mimicked the facial expressions and behavioral mannerisms of the confederate. Participants smiled more when working with a confederate who smiled rather than held a neutral expression throughout the session, and participants rubbed their face or shook their foot more when working with a confederate who performed the same mannerism compared to one who performed a different mannerism.

Mimicry (behavioral matching) has both spatial and temporal elements. In the temporal domain, individuals in many instances mimic the timing of each other's movements intentionally, such as a quartet of musicians performing together in synchrony or a marching band synchronizing their movements on the field during a half-time show. Synchrony, as a form of temporal mimicry, can also occur unintentionally,

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however, such as when walking together with another individual side-by-side—in such situations, individuals can unintentionally fall into lockstep with their walking partner. Not to the same degree that troops do when they are marching together, but synchronized nonetheless.

One question is the extent to which synchrony, as a form of temporal mimicry, extends to non-temporal (spatial) aspects of an individual's movements. One well-established paradigm that has been extensively used to examine explicit (intentional) movement synchronization (and is readily adaptable to examine mimicry of both spatial and temporal elements of an individual's movements) is the synchronization-continuation tapping paradigm (Stevens 1886). In this paradigm, individuals synchronize finger taps with a rhythmic pacing stimulus (e.g., an auditory metronome) and then continue tapping at the same rate (tempo) once the stimulus stops. For most of the studies using this paradigm, the primary data of interest have been time series of tap onsets and the associated sequence of inter-tap-intervals, which have then been used to make inferences about the nature of the underlying timing processes (e.g. Ivry and Corcos 1993; Ivry and Hazeltine 1995; Wing and Kristofferson 1973a,b). See Repp (2005), Repp and Su (2013) for two comprehensive reviews of the tapping literature.

The continuous movement dynamics of individuals' synchronize-continue tapping performance have rarely been investigated and only one previous study that we are aware of has considered whether individuals mimic spatial elements of the task. One study that investigated how movement dynamics in synchronize-continue tapping may relate to underlying timing processes is Balasubramaniam et al. (2004) who had participants "tap" their index finger in the air and instructed them to time the peaks of finger flexion or extension on or off the beat of a metronome. The authors then decomposed each inter-tap-interval into two sub-phases, flexion and extension, and analyzed the velocity profiles of each phase. They found that finger movements were "asymmetrical," where flexion and extension had different mean durations and peak velocities that depended on what was the temporal target. Specifically, the movement toward the temporal target, whether flexion or extension phase, was faster than the slower movement away from the temporal target. Moreover, they found that larger asymmetries in flexion–extension trajectories (in term of velocity and mean duration) were associated with smaller temporal asynchronies. The authors interpret this finding as evidence that the two phases of tap motion serve different temporal functions, with faster "to target" movements providing perceptual information for phase correction and slower "away from target" movements subserving period correction processes. Overall, this study highlights the importance of understanding how movement trajectories influence timing performance (i.e., more than just

tap onsets matter for understanding synchronize-continue tapping performance).

With respect to the question of mimicry, Hove and Keller (2010) had people synchronize finger tapping to one of two visual metronomes displaying either high or low amplitude finger tapping on the computer screen. The high amplitude metronome flashed alternating images of a finger resting on a table and a finger raised 5 cm above it, and the low amplitude metronome alternated between images of a finger resting on a table and one raised 2 cm above it. They found that participants roughly matched their tap amplitude (the maximum height of their finger extension between taps) to that of the height of the visual metronome even though the task was simply to synchronize their tapping with the metronome and no mention was made of matching finger height. Tap amplitude was higher in the high amplitude condition compared to the low amplitude condition. This study demonstrated that individuals implicitly mimic spatial elements of a visual metronome while performing an explicit temporal synchronization task.

The present study considers whether the same kind of implicit spatial mimicry extends to two individuals when they tap in synchrony together—a much more dynamic and interactive process than a single individual tapping in synchrony with an image of a finger moving up and down at one of two heights with perfectly isochronous timing. If people, in this more dynamic situation, mimic elements of each other's movements (e.g., how high they move their finger off the table during tapping together), this would (1) demonstrate that people are sensitive to (and unintentionally match) very subtle aspects of each other's movements that are not directly related to synchronizing the timing of their finger taps and (2) highlight a role for spatial information in a task where inferences about basic timing processes have been previously based almost exclusively on the time series of tap onsets only. With respect to the latter, the question of spatial mimicry and its potential role in timing processes builds on the work of Balasubramaniam et al. (2004).

Toward these ends, two experiments were conducted that examined continuous movement dynamics of interpersonal synchronization using a synchronize-continue tapping paradigm in which pairs of individuals tapped with their index finger on a table in synchrony with an auditory metronome and then continued tapping at the same rate (tempo) when the metronome stopped. Pairs of individuals either tapped together with the instruction to maintain synchrony with each other (interpersonal tapping) or tapped alone (solo tapping). Solo tapping conditions occurred with their tapping partner either present in the testing room (Experiment 1) or absent (Experiment 2). Continuous motion tracking was used to examine both the spatial and temporal elements of movement dynamics during task performance.

Based on Hove and Keller (2010), if individuals mimic each other's movement dynamics during interpersonal synchronization, then tapping amplitudes should be correlated during paired synchronize-continue tapping, but not during solo tapping. However, we also consider that it is possible that a participant simply viewing their tapping partner during solo tapping blocks might also lead to that participant mimicking tapping amplitudes during solo tapping blocks. One reason to suspect this possibility is that previous work has shown that unintentional synchrony is influenced by visual information. In a study by Richardson and colleagues, two individuals sat side-by-side in rocking chairs and rocked at their own (independent) preferred paces, either fixating on a target on the armrest of the other person's rocking chair or a target on the wall. Results showed that people unintentionally synchronized rocking when they were afforded visual information about the other's movements (Richardson et al. 2007). If in the present study, participants similarly use visual information alone to mimic spatial elements of each other's movements, then we might also expect to see correlated tapping amplitudes during solo tapping blocks in Experiment 1 (where the two participants were able to watch each other tap in the solo blocks), but not in Experiment 2 (where the non-tapping participant was out of the room during the solo blocks).

The present study additionally considered mimicry of another aspect of tapping performance, specifically how long each participant leaves their finger on the table in between taps (dwell time). The motivation for examining dwell time stemmed from the observation of large individual differences in how long people tend to leave their finger on the table in between taps, and thus we saw dwell time as another aspect of tapping that participants might mimic in interpersonal tapping conditions. Standard measures of synchronize-continue tapping performance (e.g., mean and standard deviations of produced inter-tap-intervals) were also examined to consider overall differences in timing performance between solo and interpersonal tapping conditions.

A final consideration of the present study was based on previous work that behavioral mimicry is associated with better rapport between individuals (Hove and Risen 2009; Tickle-Degnen and Rosenthal 1987). Most closely related to the present study, Hove and Risen (2009) had participants tap with an experimenter who either tapped in synchrony or out of synchrony with the participant. They found that higher likability ratings for the experimenter were correlated with smaller interpersonal asynchronies and participants in the synchronous group liked the experimenter significantly more than participants in the asynchronous group, indicating that greater interpersonal synchrony caused greater affiliation. To revisit this observation, all participants in the present study completed an affiliation survey at the end of the study that

included questions such as: “How likeable was the person you tapped with today?” and “How much would you like to participate in another study with the person you tapped with today?”.

Experiment 1

Methods

Participants and design

Sixty-four undergraduates (54 female, 18–27 years old; $M = 19.1$ yrs, $SD = 1.6$) completed the experiment in return for credit in an undergraduate psychology course. All participants self-reported normal hearing and varied in the number of years of formal music training (0–13 years, $M = 2.8$, $SD = 3.4$), and eight participants were left-handed. The design of the experiment was a 2 (LH Tapper vs. RH Tapper) X 2 (Tapping Condition: Solo vs. Paired) mixed-factorial design. Participants were assigned to pairs based on the time of their appointment and were randomly assigned to be LH Tapper (Left Hand Tapper) or RH Tapper (Right Hand Tapper). Participants sat on opposite sides of the table, with the LH Tapper tapping with their left hand and RH Tapper tapping with their right hand, regardless of their handedness, with participants' non-tapping hand resting on their lap throughout the study (see Fig. 1). The Institutional Review Board at Michigan State University approved this study and all participants provided informed consent.

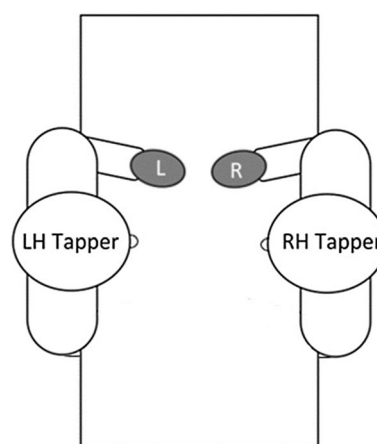


Fig. 1 Schematic of tapping dyad sitting across from each other with their hands atop a wooden table. One participant tapped with their right hand (RH Tapper) and the other tapped with their left hand (LH Tapper) throughout the experiment. Participants sat across from each other so that the tapping hands of LH Tapper and RH Tapper aligned. Participants' non-tapping hands rested on their laps

Stimuli and apparatus

The synchronization pacing signal consisted of sixteen 50-ms 440-Hz sine tones followed by 500 ms of silence to create a sequence of 15 equal inter-onset-intervals (IOIs) corresponding to a target inter-tap-interval (ITI) of 550 ms. A longer, 200-ms 440-Hz sine tone, signaled participants to stop tapping after ~60 produced ITIs (15 for the synchronization phase and 45 for the continuation phase). Pacing tones were presented over Sennheiser HD 280 Pro headphones at a comfortable listening level. We collected continuous movement data using an Ascension trakSTAR electromagnetic motion tracking system (trakSTAR by Ascension Technology Corp, Shelburne, VT, USA), with a sensor attached to the distal phalanx of the each participant's tapping index finger, recording *x*, *y*, and *z* coordinates with a 200 Hz sampling rate.

Procedure

Participants completed the experiment in dyads (pairs). Prior to receiving specific instructions, participants sat, facing each other, at a table across from each other, placing their tapping hand comfortably atop the table and with their non-tapping hand resting on their lap (see Fig. 1). Based on which side of the table they sat at, one participant was the 'LH Tapper' (tapping with their left hand) and one participant was the 'RH Tapper' (tapping with their right hand). The instructions to participants were as follows: "In this study, you will be asked to tap with your index finger on the table. Your job is to tap in synchrony with an auditory metronome (rhythm) and then continue to tap at the same pace once the sounds stop. One of you will tap with your LEFT index finger, and the other will tap with your RIGHT index finger throughout the study. On some trials, you will tap alone, and on other trials, you will tap at the same time together with the person sitting across from you. When tapping together, please do your best to synchronize with the other person you are tapping with throughout the trial." After receiving instructions, participants were encouraged to ask questions to make sure they understood the task and the experimenter checked that participants' non-tapping hand was on their lap and that they were only tapping with the hand on the table.

The overall structure of the experiment consisted of a block of five solo tapping trials followed by a block of five paired tapping trials (where participants tapped together), followed by a second block of five solo tapping trials, with brief rest breaks between blocks. For the solo tapping trials, each participant (in turn) tapped in synchrony with 16 isochronous 'pacing' tones with a target 550-ms inter-tap-interval and then continued tapping at the same tempo once the pacing tones stopped until the end of the trial. During solo

tapping, the non-tapping participant rested the hand with the sensor attached atop the table with the explicit instruction to not tap or move along with the other participant. For the paired tapping trials, participants synchronized together to the pacing signal during the synchronization phase and then continued tapping together in synchrony once the pacing tones stopped until the end of the trial. A longer 200-ms tone and an image of a stop sign presented on the computer monitor signaled the end of each trial. Each trial was 40 s in duration, resulting in the 16 synchronization taps (15 synchronization intervals) and at least 46 continuation taps (45 continuation intervals) for each participant.

The order of solo tapping (starting with either LH Tapper or RH Tapper) was counterbalanced within participant pairs, such that the participant who tapped first during the first solo tapping block (before the paired tapping) always tapped second during the second solo tapping block (after the paired tapping). Additionally, the order of solo tapping in the first solo tapping block (LH Tapper vs RH Tapper) was counterbalanced between participant pairs, such that the first solo tapper in the first solo tapping block alternated between the LH Tapper and RH Tapper across pairs. After completing the solo and paired synchronize-continue tapping portions of the experiment, participants each completed a short survey of perceived affiliation with their assigned partner, a short demographics survey, and a short handedness survey. The entire experiment lasted ~60 min.

Data analysis

The primary dependent measures of interest were mean tap peak amplitude (representing the maximum height of finger movement), mean dwell-time (duration of finger contact with table following a tap onset) (see Fig. 2), mean produced

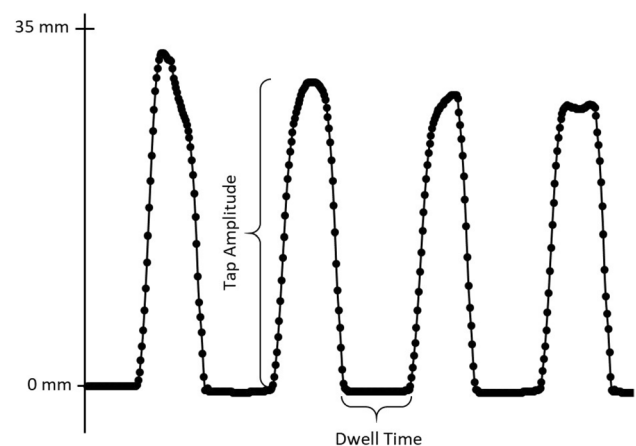


Fig. 2 Continuous amplitude profile from one finger showing tap amplitude and dwell time for several taps. Each point represents vertical finger position sampled every 5 ms

inter-tap-interval (tapping accuracy), and standard deviation of produced inter-tap-intervals (tapping variability), which were calculated from the time series of taps extracted from the continuous motion data for each trial using a custom MATLAB program. Prior to all formal analyses, tapping data were first filtered to remove any false taps (e.g., finger bounces) identified by the ‘findpeaks’ function in MATLAB. This was done automatically by removing identified taps that were within 50 ms of the preceding identified tap in the time series—a method we developed based on visual inspection of the continuous movement data and previous tapping studies (Helmuth and Ivry 1996). For the calculation of mean and standard deviation of produced inter-tap-intervals, ITIs that were less than 50% of the target ITI or 50% greater than the target ITI were considered errors and eliminated from the analysis following Helmuth and Ivry (1996). To calculate the standard deviation of produced inter-tap-intervals (ITIs), the sequence of ITIs was first linearly de-trended to remove any systematic drift (e.g., gradual speeding up or slowing down), which can inflate estimates of variability due to violation of the assumption of a stationary mean. For the de-trending process, a linear regression line was fit to the sequence of continuation inter-tap-intervals for each trial and SD was then calculated using the residuals rather than the raw ITI sequence (i.e., after any drift was removed) (Helmuth and Ivry 1996; Ivry et al. 1988). Estimates of mean and standard deviation of ITIs were averaged over each block of 5 trials (Helmuth and Ivry 1996; Ivry et al. 1988; Keele et al. 1985).

Because trial duration was fixed at 40 s, some trials had more than 46 continuation taps (45 inter-tap-intervals) depending on individual variation in tapping speed during the continuation phase. Data analyses reported below only considered the first 46 continuation taps (45 inter-tap-intervals) for each participant. Any trial that contained fewer than 45-produced continuation ITIs was eliminated (1.8% of trials).

Results

Figure 3 shows representative examples of movement trajectories for one dyad during a solo synchronize-continue tapping trial (top panel) and a paired synchronize-continue tapping trial (bottom panel). Finger tapping amplitude is represented on the y-axis as a function of time in 5-ms increments. Informal visual inspection of this sample dyad comparing paired and solo tapping suggests greater similarity of the dyad’s movement trajectories during paired tapping than solo tapping. Figures 4 and 5 show the formal relation between tap amplitudes and dwell times, respectively, for each dyad during solo-pre (Panel A), paired (Panel B), and solo-post (Panel C) tapping conditions. Consistent with the mimicry hypothesis, tapping amplitudes and dwell times were positively correlated during paired tapping (Tapping Amplitude: $r(30)=0.83$, $p<0.001$, 95% CI [0.62, 1.04]; Dwell Time: $r(30)=0.45$, $p=0.009$, 95% CI [0.12, 0.78]). During solo tapping conditions, tapping amplitudes were positively correlated, but only significantly so in the

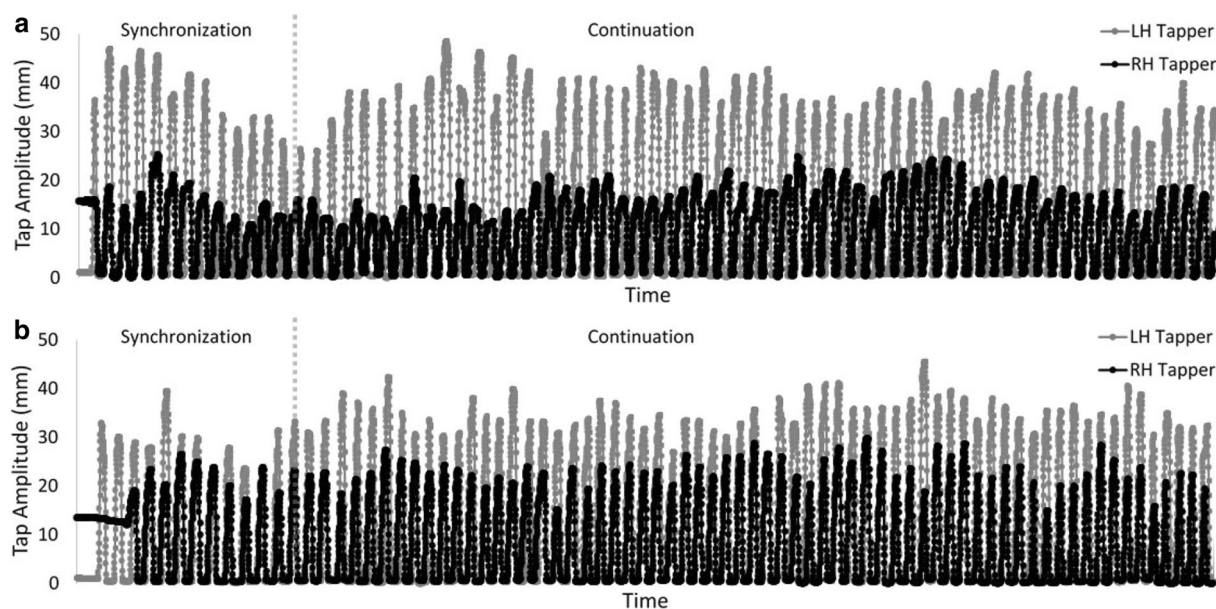


Fig. 3 Representative example of continuous amplitude profiles from one tapping dyad showing one solo trial (**a**) and one paired trial (**b**). LH Tapper is indicated in grey and RH Tapper is indicated in black. Y-axis position of 0 mm corresponds to the table top. The region to the left of the dotted line is the synchronization phase, and the region

to the right of the dotted line is the continuation phase. Although the LH Tapper had higher amplitude than RH Tapper overall in this example dyad, there was not a consistent difference between mean LH and RH Tapper amplitude

solo-post condition (solo-pre, $r(30)=0.30$, $p=0.10$, 95% CI $[-0.06, 0.66]$; solo post, $r(30)=0.62$, $p<0.001$, 95% CI $[0.33, 0.91]$). Dwell times during solo tapping were not reliably correlated for either the solo-pre, $r(30)=0.34$, $p=0.06$, 95% CI $[-0.01, 0.69]$ or solo-post conditions, $r(30)=-0.05$, $p=0.79$, 95% CI $[-0.42, 0.32]$.

Next, we considered potential differences in tapping accuracy (i.e., how closely participants matched the target 550 ms ITI) during solo and paired tapping. Figure 6 shows the sequence of mean produced inter-tap-intervals (ITIs) during the synchronization and continuation phases of the tapping task for the solo and paired tapping. A visual inspection of Fig. 6 illustrates two general phenomena of interest.

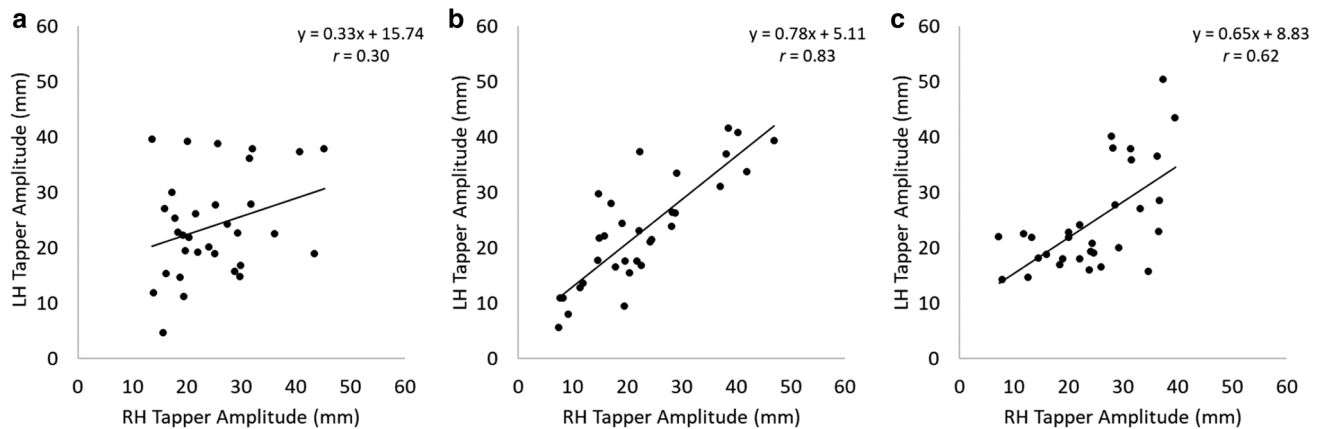


Fig. 4 Experiment 1: Relation between maximum tap amplitude (finger height) for each dyad during solo-pre (a), paired (b) and solo-post (c) tapping conditions

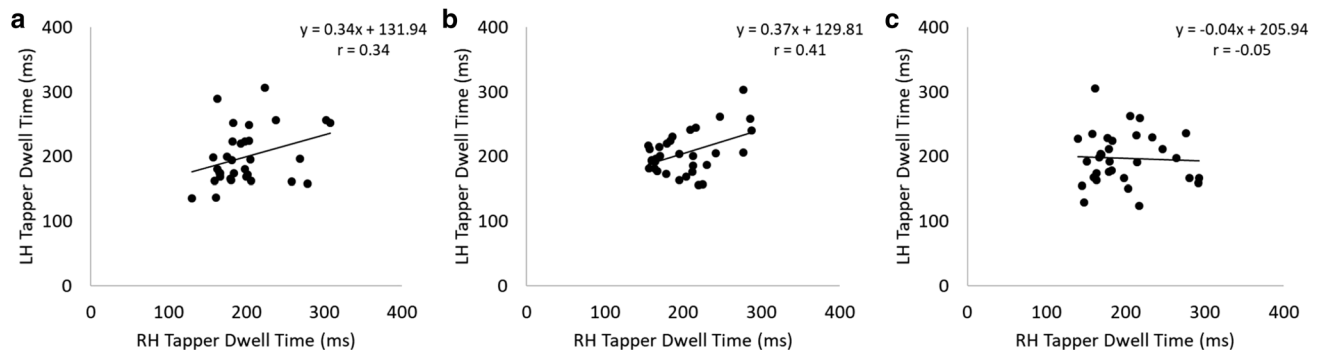


Fig. 5 Experiment 1: Relation between tap dwell times for each dyad during solo-pre (Panel a), paired (b) and solo-post (c) tapping conditions

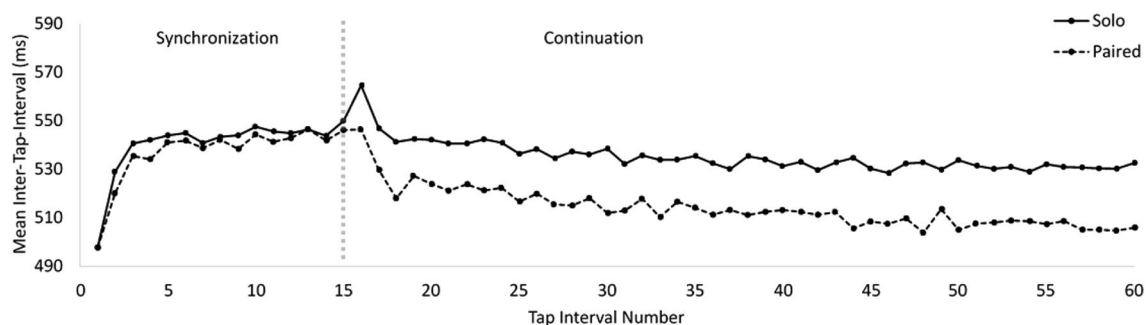


Fig. 6 Experiment 1: Mean produced inter-tap-intervals during synchronization and continuation during solo tapping (solid lines) and paired tapping (dashed line). The region to the left of the vertical dot-

ted line is the synchronization phase, and the region to the right of the dotted line is the continuation phase

First, during solo tapping, a hesitation consistently occurred at the end of the synchronization phase that is not present during paired tapping. Second, during paired tapping, participants sped up their tapping during the continuation phase. With respect to the latter phenomenon, mean ITIs during the synchronization phase did not differ between solo and paired tapping condition (solo: $M = 539.7$, 95% CI, 536.1–543.3; paired, $M = 536.4$, 95% CI, 532.2–540.5), but during the continuation phase mean ITIs were significantly shorter for paired tapping ($M = 507.1$, 95% CI, 499.2 – 515.0) than for solo tapping ($M = 527.3$, 95% CI, 520.0 – 534.5). Consistent with this interpretation, a 2 (LH Tapper vs. RH Tapper) X 2 (Tapping Condition: Solo vs. Paired) mixed-measures ANOVA on mean continuation ITIs revealed a significant main effect of Tapping Condition, $F(1,62) = 28.90$, $p < 0.001$, $\eta_p^2 = 0.32$, but no main effect of LH vs. RH Tapper $F(1,62) = 0.13$, $p = 0.72$, $\eta_p^2 = 0.002$, or interaction between Tapper and Tapping Condition, $F(1,62) = 0.22$, $p = 0.64$, $\eta_p^2 = 0.004$.

Next, we compared tapping precision (variability) across conditions. Standard deviations of mean ITIs were similar for solo tapping ($M = 50.6$, 95% CI, 47.3–54.0) and paired tapping ($M = 49.3$, 95% CI, 46.7–52.0) and for the LH Tapper ($M = 52.3$, 95% CI, 48.4–56.2) and the RH Tapper ($M = 47.6$, 95% CI, 43.7–51.6). A 2 (LH Tapper vs. RH Tapper) X 2 (Tapping Condition: Solo vs. Paired) mixed-measures ANOVA on SD of produced continuation ITIs revealed no effect of Tapper, $F(1,62) = 2.89$, $p = 0.10$, $\eta_p^2 = 0.04$, no effect of Tapping Condition, $F(1, 62) = 1.08$, $p = 0.30$, $\eta_p^2 = 0.02$, and no interaction between Tapper and Tapping Condition, $F(1,62) = 2.09$, $p = 0.15$, $\eta_p^2 = 0.03$.

Finally, we considered whether the fact that there were some left-handed individuals in the sample affected any of the mimicry results by calculating correlations between dyads' tapping amplitudes and dwell times without left-handed participants ($n = 8$ dyads removed). This smaller sample of only right-handed participants contained 24 dyads, each composed of one participant tapping with their non-dominant hand (the LH Tapper) and one participant tapping with their dominant hand (the RH Tapper). The pattern of correlations for right-handers only was the same as for the full sample. The one minor exception was that the correlation of dwell times between participants during solo-pre trials, which was only marginally significant in the full sample, $r(30) = 0.34$, $p = 0.06$, 95% CI [–0.01, 0.69], was significant in the right-hand-only sample, $r(22) = 0.55$, $p < 0.01$, 95% CI [0.17, 0.92]. Overall, these results suggest that handedness plays a minimal role in the observed mimicry.

Discussion

The results of Experiment 1 provide evidence of behavioral mimicry. Participants performing an interpersonal

synchronize-continue tapping task mimicked subtle aspects of each other's movements not directly related to the performance of the task. In addition to matching each other's inter-tap-intervals (ITIs), participant dyads also mimicked each other's tap amplitudes (how high they lifted their finger off the table) and dwell times (how long they left their finger on the table in between taps). To a lesser degree, mimicry of tap amplitudes was also found for solo tapping trials, particularly for solo-post trials that followed paired tapping. A consideration of handedness revealed the same pattern of results for the analysis of the full sample, which included some left-handed participants, as with the reduced sample of dyads consisting of only right-handed participants.

At first glance, the reliable mimicry of tapping amplitude during the solo-post tapping trials is somewhat surprising. There are at least two potential explanations for this effect, however. One possibility is that participants continue tapping at their adjusted amplitude for some amount of time following interpersonal tapping. Another explanation is that simply watching each other tap during solo blocks is sufficient for some mimicry to occur and that this effect is greater in the solo-post block than in the solo-pre block as participants have simply had longer to watch each other tap. The weak correlation between amplitudes in solo tapping trials preceding interpersonal tapping could have been a result of the second solo tapper always having observed the first person tap and mimicking their tap amplitude. A combination of these two explanations could account for the larger correlation in unpaired tapping amplitude during the solo-post compared to solo-pre block.

To distinguish these possibilities, we conducted a second experiment that was identical to Experiment 1, except that participants performed solo tapping blocks without their tapping partner in the room. We also tested only right-handed participants to control for handedness. If the carryover effect on tap amplitude was a result of the persistence of dynamic changes due to mimicry after the interpersonal block, then we would expect to see a correlation between tappers' amplitudes in the solo-post tapping block even if the other tapper is not in the room. If the carryover effect was a result of sensitivity to the other tapper's spatial dynamics when observing them tap during solo blocks, then we would expect to see no correlation between tappers' amplitudes in the solo-post tapping blocks when tappers are not in the same room. Because previous research has shown that better interpersonal synchrony is related to better interpersonal rapport (Hove and Risen 2009), in Experiment 2 we also examined the relationship between interpersonal synchrony (mean and SD of phase differences between LH and RH tapper) and affiliation between each dyad member.

Experiment 2

Methods

Participants and design

Sixty-four undergraduate students (45 female, 8–22 years old; $M = 18.9$ yrs, $SD = 1.1$) completed the experiment in return for credit in an undergraduate psychology course. All participants were right-handed and self-reported normal hearing. Participants varied in number of years of formal music training (0–12 years, $M = 2.9$, $SD = 3.5$). The experiment implemented a 2 (Tapper: LH Tapper vs. RH Tapper) X 2 (Tapping Condition: Solo vs. Paired) mixed-factorial design. Tapper was a between-subject variable, while the tapping condition was a within-subject variable.

Stimuli and apparatus

Identical to Experiment 1.

Procedure

The procedure was identical to Experiment 1, except that during solo tapping blocks, the non-tapping participant left the room and could not see (nor hear) the tapping participants while they were tapping.

Data analysis

Identical to Experiment 1.

Results

Figures 7 and 8 show the relation between tap amplitudes and dwell times, respectively, for each dyad during solo-pre (Panel A), paired (Panel B), and solo-post (Panel C) tapping conditions. Consistent with the mimicry hypothesis, tapping amplitudes were highly positively correlated during paired tapping, $r(30) = 0.58$, $p = 0.001$, 95% CI [0.27, 0.88]. However, in contrast to Experiment 1 (where during solo tapping trials, the non-tapping participant of each

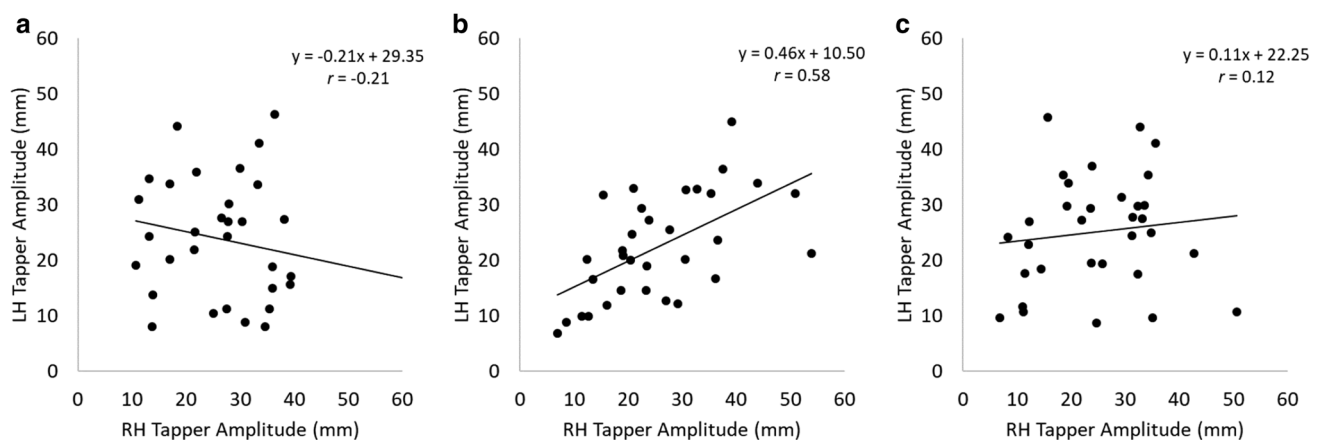


Fig. 7 Experiment 2: Relation between maximum tap amplitude (finger height) for each dyad during solo-pre (a), paired (b) and solo-post (c) tapping conditions

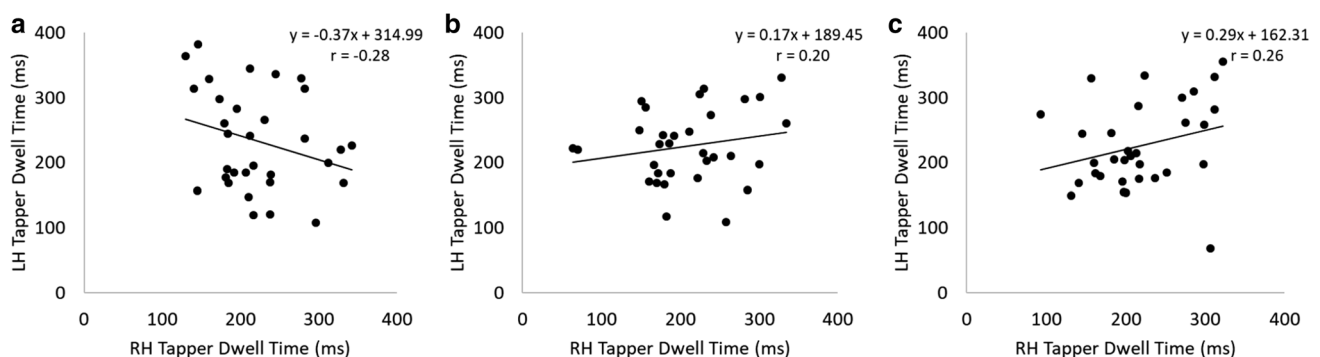


Fig. 8 Experiment 2: Relation between tap dwell times for each dyad during solo-pre (a), paired (b) and solo-post (c) tapping conditions

dyad could see the tapping participant), tapping amplitudes were not correlated during solo-pre, $r(30) = -0.21$, $p = 0.25$, 95% CI $[-0.57, 0.15]$, and solo-post, $r(30) = 0.12$, $p = 0.51$, 95% CI $[-0.25, 0.49]$, tapping conditions. Thus, eliminating visual information provided to the non-tapping participant about the tapping participant's movement eliminated the positive correlation observed between tapping amplitudes during solo-pre and solo-post tapping blocks in Experiment 1. Dwell times were not reliably correlated during solo-pre, $r(30) = -0.28$, $p = 0.12$, 95% CI $[-0.64, 0.08]$, solo-post, $r(30) = 0.26$, $p = 0.14$, 95% CI $[-0.10, 0.62]$, or paired tapping, $r(30) = 0.20$, $p = 0.28$, 95% CI $[-0.17, 0.56]$.

Table 1 unpacks tapping amplitude correlations by trial for Experiments 1 and 2. Here, we considered the consistency of mimicry across trials and the possibility that carryover of mimicry from paired to solo-post tapping for tapping amplitude was present in early but not later solo-post trials. Correlations between dyads' tapping amplitudes by trial show that carryover observed in Experiment 1 persisted across all solo-post trials. Amplitude mimicry did not decay over the course of the solo-post block. In Experiment 2, there was no mimicry observed in any solo-post trials,

Table 1 Pearson correlation, r , between maximum tap amplitude for members of each dyad for solo-pre, paired, and solo-post tapping conditions in Experiments 1 and 2

	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
Experiment 1					
Solo-Pre	0.42*	0.32	0.34	0.21	0.20
Paired	0.72**	0.82**	0.85**	0.89**	0.86**
Solo-Post	0.62**	0.58**	0.61**	0.63**	0.49**
Experiment 2					
Solo-Pre	-0.33	-0.19	-0.15	-0.07	-0.16
Paired	0.57**	0.62**	0.56**	0.56**	0.56**
Solo-Post	0.13	0.11	0.17	0.11	0.05

*Indicates significance at the $p = 0.05$ level, ** indicates significance at the $p = 0.01$ level

indicating that the lack of carryover effects in Experiment 2 was not due to averaging over trials in the solo-post block.

Next, we considered potential differences in tapping accuracy (i.e., how closely participants matched the target 550 ms ITI) during solo and paired tapping. Figure 9 shows the sequence of mean produced inter-tap-intervals (ITIs) during the synchronization and continuation phases of the tapping task for the solo and paired tapping. The same general pattern was observed in Experiment 2 as in Experiment 1. First, during solo tapping, a hesitation consistently occurred at the end of the synchronization phase that was not present during paired tapping. Second, during paired tapping, participants sped up their tapping during the continuation phase. As found in Experiment 1, mean produced inter-tap-intervals did not differ between solo and paired tapping during the synchronization phase (solo: $M = 543.3$, 95% CI, 541.3–545.2; paired, $M = 541.2$, 95% CI, 539.2–543.3), but mean ITIs during the continuation phase were shorter (i.e., participants tapped faster) during paired tapping ($M = 519.5$, 95% CI, 512.1–526.9) than during solo tapping ($M = 540.8$, 95% CI, 535.6–545.9). A 2 (LH Tapper vs. RH Tapper) X 2 (Tapping Condition: Solo vs. Paired) mixed-measures ANOVA on mean produced continuation ITIs showed a main effect of Tapping Condition, $F(2,62) = 34.18$, $p < 0.001$, $\eta_p^2 = 0.36$, but no main effect of Tapper, $F(1,62) = 0.27$, $p = 0.69$, $\eta_p^2 = 0.004$, or interaction between Tapper and Tapping Condition, $F(1,62) = 0.31$, $p = 0.58$, $\eta_p^2 = 0.005$.

Next, we compared tapping precision (variability) across conditions. A 2 (LH Tapper vs. RH Tapper) X 2 (Tapping Condition: Solo vs. Paired) mixed-measures ANOVA on SD of produced continuation ITIs revealed an effect of Tapping Condition, $F(1, 62) = 7.81$, $p = 0.007$, $\eta_p^2 = 0.11$, no effect of Tapper, $F(1,62) = 0.20$, $p = 0.65$, $\eta_p^2 = 0.003$, and no interaction between Tapper and Tapping Condition, $F(1,62) = 0.93$, $p = 0.34$, $\eta_p^2 = 0.01$. Standard deviations of mean ITIs were slightly lower (better) for solo tapping ($M = 43.7$, 95% CI, 41.0–46.4) than paired tapping

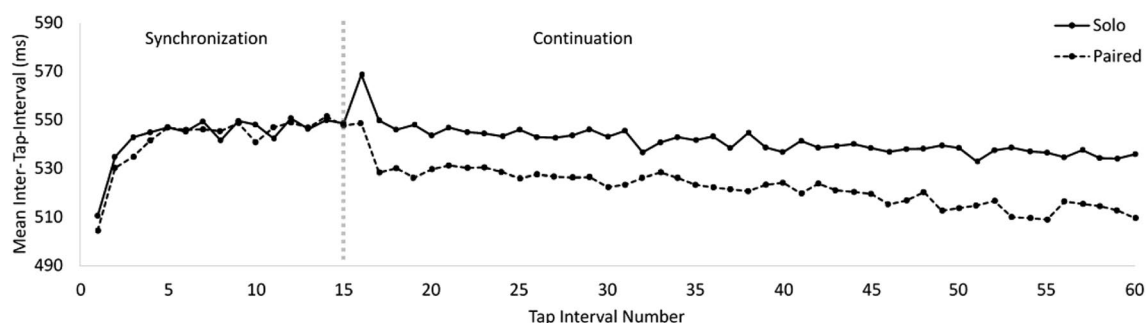


Fig. 9 Experiment 2: Mean produced inter-tap-intervals for synchronization and continuation during solo tapping (solid line) and paired tapping (dashed line). The region to the left of the vertical dotted line

is the synchronization phase, and the region to the right of the dotted line is the continuation phase

($M=46.7$, 95% CI, 44.0–49.5), but there was no difference between the LH Tapper ($M=45.8$, 95% CI, 42.2–49.3) and RH Tapper ($M=44.7$, 95% CI, 41.1–48.2).

Finally, to evaluate a potential relationship between interpersonal synchrony and affiliation, we examined correlations between two measures of interpersonal synchrony between dyad members and their affiliation ratings (Table 2). To measure interpersonal synchrony, both mean and standard deviation (SD) of phase differences between continuation tap onsets for the LH and RH tapper in each dyad (RH Tapper phase–LH Tapper phase). For these measures, smaller mean phase differences indicate greater average interpersonal synchrony, whereas smaller phase difference variability (SD) indicates more consistency of interpersonal synchrony. Neither measure was found to be reliably associated with participants' feeling of affiliation with the other member of the dyad (all p 's > 0.07).

Discussion

Results of Experiment 2 replicated the amplitude mimicry but not the dwell time mimicry results of Experiment 1. As found in Experiment 1, when participants in each dyad tap in synchrony together (a temporal task), they mimic how high they move their finger off the table (a spatial element of their movement). Dwell times were also positively correlated in Experiment 2, but not significantly so. Moreover, the lack of a correlation in tapping amplitudes in solo-pre and solo-post trials in Experiment 2 suggests that the solo-pre and solo-post amplitude correlations observed in Experiment 1 were due to the non-tapping participants being able to see the tapping participants movements during solo tapping trials. Experiment 2 also replicated the 'hesitation' and 'speeding up' phenomena observed in Experiment 1, whereby participants during solo tapping tended to hesitate at the end of the synchronization phase during solo tapping but not paired tapping and tended to speed up more during the continuation phase during paired tapping, but not solo tapping. Finally, our results did not replicate those of Hove and Risen (2009)

in that we did not find any reliable correlations between measures of interpersonal synchrony and dyad members' feelings of affiliation.

General discussion

Two experiments examined implicit mimicry of movement dynamics during an interpersonal synchronize-continue tapping task. In both experiments, pairs of participants tapped their finger on a table in synchrony with an isochronous auditory pacing stimulus and then continued tapping at the same rate after the pacing signal stopped. Tapping occurred either alone with one hand (solo) or together with the instruction to tap in synchrony with each other (paired) while seated across the table facing each other. Of particular interest was whether participants would implicitly mimic how high they moved their finger off the table (tap amplitude) and how long they kept their finger in contact with the table between taps (dwell time) while tapping in synchrony together. The two experiments differed in whether during the solo tapping blocks the non-tapping participant was present in the room and could see the tapping participant (Experiment 1) or not (Experiment 2).

The results revealed four main findings: first, individuals implicitly mimicked the height (amplitude) of each other's finger movements during paired tapping (Experiments 1 and 2), and to a lesser degree during solo tapping when the non-tapping participant had been able to watch the tapping participant's finger movements (Experiment 1). Second, supporting that the mimicry observed during the solo tapping blocks was due to implicit use of the visual information (non-tappers being able to see tappers), having the non-tapping participant in a different room during the pre- and post-solo tapping blocks (Experiment 2) eliminated the implicit spatial mimicry observed during solo tapping in Experiment 1. Third, during paired tapping in Experiment 1, participants also mimicked each other's dwell time (how long they kept their finger on the table between taps), but this finding was only observed in Experiment 1, and was not strong. Fourth, separate from the observed implicit mimicry, there was a consistent tendency in terms of the sequence of produced ITIs in Experiments 1 and 2 for participants to hesitate during the end of the synchronization phase of solo tapping but not paired tapping, while, conversely, to speed up during the continuation phase of paired tapping but not solo tapping.

Overall, the mimicry results for dwell times (how long participants left their finger on the table in between taps) were less consistent than the mimicry results for amplitude (finger height). Dwell times were reliably correlated between participants during paired tapping in Experiment 1, but not in Experiment 2. Moreover, it is notable that correlations in Experiment 2 were overall weaker than in

Table 2 Pearson correlation values for interpersonal continuation tapping synchrony measures and affiliation ratings made by each dyad member about the other in Experiment 2

	Mean LH tapper affiliation rating	Mean RH tapper affiliation rating
Mean phase difference	0.33	0.14
Mean phase difference variability (SD)	0.03	– 0.15

Likability ratings were made by the LH Tapper about the RH Tapper and vice-versa in response to the question "how likeable was the person you were tapping with today?" Overall, interpersonal synchrony was not significantly related to dyad affiliation

Experiment 1. With respect to dwell times, the dwell time correlation between participants during paired tapping was still positive in Experiment 2, but no longer significant. A possible reason for the non-significant dwell time correlation and, more broadly, for the overall weaker correlations in Experiment 2 than in Experiment 1 may be that the non-tapping member of each member was not present in the room during solo tapping conditions in Experiment 2, but was present in Experiment 1. Specifically, visual access to the other member of the dyad during solo-pre tapping in Experiment 1 may have afforded longer time for participants to become familiar each other's tapping. This view fits with previous research that has shown that simply watching a motor action being executed can facilitate learning and performing that action (for reviews see: Heyes 2001; Wulf and Prinz 2001; Blakemore and Frith 2005). Thus, being present in the room during the solo-pre tapping would potentially strengthen mimicry in the paired condition for Experiment 1 compared to Experiment 2.

One question that emerges from this work is whether the observed mimicry was unintentional. That is, although participants were only instructed to maintain synchrony with their tapping partner, which focuses their attention on timing, perhaps participants also explicitly tried to match other aspects of their movement dynamics, such as how high they moved their finger off the table. Although we cannot completely rule this possibility out, it seems unlikely for a couple of reasons. First, unlike when dancing, tapping with an index finger is a subtle movement and associated small differences in maximum finger height (on a millimeter scale) do not generally draw attention. Second, in line with the first observation, when we examined participants' responses to a survey given to all participants at the end of the experiment, which includes questions about describing any strategy they used to do the task, no participant mentioned matching finger height. Rather, the strategies (when reported) tended to emphasize, as expected, the temporal aspects of the movement—beat and rhythm (e.g., “On the partner tapping I followed and lead the beat of tapping”).

Overall, the observed mimicry of individuals' finger height (and to a lesser extent dwell times) during interpersonal synchronize-continue tapping extends the work of Hove and Keller (2010) who found that single individuals performing a synchronization task implicitly matched the maximum height of their finger movement with the height of the image of a finger rhythmically moving up and down on a computer screen at one of two heights. Here, the present study is the first to show similar implicit spatial mimicry *between* individuals in a more interactive interpersonal synchronize-continue tapping task where pairs of individuals synchronized their tapping together. These results further highlight the potential for spatial information to

influence the performance of a task that previous work has primarily viewed as a purely temporal task.

The finding that individuals within a dyad were sensitive to each other's movement dynamics when simply observing the other's movements is consistent with previous research demonstrating that visual feedback is an important source of information for spatiotemporal synchronization in absence of salient auditory feedback (Ceux et al. 2003; Richardson et al. 2005; Schmidt and O'Brien 1997). Results here showed that following the interpersonal synchronization episode, individuals continued to mimic the maximum height of each other's finger movements during the solo-post tapping condition, but only when the solo-post condition occurred in the presence of their tapping partner (Experiment 1, but not Experiment 2). Amplitude matching in this condition indicated that participants' observation of their partner tapping influenced their own solo tapping dynamics.

The present study revealing contributions of visual feedback to interpersonal synchronization in a rhythmic tapping task fits within a broader body of work that has investigated, using a variety of paradigms, (1) the role of visual information and feedback to movement coordination (e.g., Wishart et al. 2002; Ceux et al. 2003) and (2) the relation between perception and action during coordinated and imitative movement (Georgiou et al. 2007; Rens and Davare 2019). Wishart et al. (2002) found that spatiotemporal coordination of bimanual movements was enhanced when augmented visual feedback was provided concurrently with the task in ‘real time’ compared to only terminal feedback at the end of each trial. Participants more successfully learned novel, bimanual 90-degree relative phase movements when they could see the results of their hand motions projected on a computer monitor as they performed the task than when the resultant motion tracks appeared after the trial. Similarly, Ceux et al. (2003) investigated the effect of visual feedback on coordination performance. Participants moved a lever from side-to-side in synchrony with the image of a green rectangle on a screen and were either provided augmented visual feedback of horizontal movement trajectory (from a laser pointer mounted on the lever), or no augmented feedback (no laser light). Participants demonstrated better spatial coordination with augmented feedback than without it. The difference between produced movement amplitude and stimulus amplitude was smaller in the feedback condition compared to no feedback. This finding, also shown with a pendulum-swinging task (Schmidt and O'Brien 1997), provides evidence that both spatial and temporal characteristics of synchronization are facilitated by visual feedback.

Two additional consistent findings of the experiments reported in this article are (1) the tendency for individuals to hesitate at the end of the synchronization phase during solo tapping, but not paired tapping and (2) the tendency to speed up tapping during the continuation phase during

paired tapping, but not during solo tapping. With respect to the former, one potential explanation of the hesitation observed during the end of the synchronization phase is that the external pacing tones drive movement and when they stop, individuals must continue tapping based on a now internally generated timing signal. No hesitation (delay) occurs in the paired tapping condition because individuals' can use each other's taps as the external pacing stimulus. This hesitation phenomenon is common during synchronize-continue tapping, but rarely formally reported; additional research is needed to better understand its basis.

One potential explanation for the tendency to speed up during the continuation phase of paired tapping, but not solo tapping, is based on the work of Konvalinka et al. (2010). According to Konvalinka et al., to achieve relative synchrony in an interpersonal synchronization task, each individual must continuously anticipate the timing of the other person's next action and adjust the timing of their own to minimize the temporal distance between the two. This leads to an interaction described by Konvalinka et al. (2010) as a "hyper-follower" relationship, in which neither member of the tapping dyad is leading the other. The "hyper-follower" relationship was posited as an explanation for greater acceleration in paired than solo continuation tapping when people could hear but not see each other tap (Okano et al. 2017). Our study extends these findings to a paired condition in which dyads visually coordinated tapping. If individuals within a dyad use tap asynchrony information to reset the phase and if resetting is biased toward the larger of two negative asynchronies, both tappers will speed up more when tapping together than when tapping alone.

Finally, with respect to the comparison of tapping variability between solo and paired tapping conditions, previous research comparing unimanual to bimanual tapping performance has revealed a bimanual advantage; when a person taps with two hands (bimanually), within-hand temporal variability is lower than when tapping with one hand (unimanually) (Helmuth and Ivry 1996). This effect has also been shown in individuals tapping with two fingers compared to one (Drewing et al. 2002) as well as one finger and one forearm compared to either effector alone (Helmuth and Ivry 1996). Helmuth and Ivry (1996) tested the bimanual advantage between individuals by having two people sit side-by-side and perform synchronize-continue tapping together (paired) and alone (solo). Similar to the present study, they did not find an interpersonal bimanual advantage, that is, within-subject temporal variability was not lower during paired than solo tapping. Our study replicated this finding with two participants sitting across from each other during tapping, an arrangement that affords more visual information transfer. Rather than finding a performance advantage for having two individuals tap together, as would be predicted if there is an

interpersonal bimanual advantage, variability of inter-tap intervals during paired tapping was the same as for solo tapping (Experiment 1) or slightly greater than for solo tapping (Experiment 2).

In conclusion, the current study demonstrates that individuals are sensitive to, and implicitly match spatiotemporal characteristics of each other's movements (maximum finger height and to a lesser extent dwell times) when synchronizing tapping, even when not instructed to do so. This is also true of individuals observing each other's movements without synchronizing and provides further evidence for perception–action coupling between individuals (Prinz 1997). Although the present study establishes that individuals modulate the spatial dynamics of their movements during the performance of what has traditionally been viewed as a purely temporal task, additional work is needed to investigate how mimicking (or simply observing) another's movements may impact inferences about underlying timing processes, which previously have been based solely on the time series of tapping onsets.

One potentially interesting line of future work is to consider mimicry during asynchronous conditions, such as dyads tapping together in anti-phase, so that taps alternate. It is unclear in that instance whether individuals would still mimic spatial characteristics of their movements if their timing is not executed synchronously. In a step in that direction, Hove and Risen (2009) compared synchronous and asynchronous rhythmic tapping conditions and observed greater feelings of affiliation in synchronous than in asynchronous conditions. Given previous work showing a relationship between mimicry and affiliation, greater spatial mimicry might also be similarly observed during synchronous (compared to asynchronous) tapping conditions. Other interesting lines of future work include more immediate tests of the direction of mimicry based on visual information by, for example, blindfolding one participant, directly manipulating spatial dynamics of the synchronization-continuation tapping task by establishing a target amplitude in addition to the target inter-tap-interval, and more directly comparing interpersonal synchronization performance between visual and auditory feedback conditions.

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Availability of data and material Data are available upon request.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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